Declarations

Journal: Journal of Neural Engineering

Manuscript: Limited-label cross-session motor-imagery EEG recalibration: spatial refitting and the added value of Euclidean alignment

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Research ethics

The present secondary analysis was exempt from additional ethical review under Article 32, item (1), of China's Measures for Ethical Review of Life Science and Medical Research Involving Human Participants (2023). It used lawfully obtained publicly available data, caused no harm to participants, and did not involve sensitive personal information or commercial interests. The original Kumar data collection reported University of Texas at Austin approval under protocol 2020-03-0073 and written informed consent. These statements refer to the original data collection. The reviewed BNCI2014-001 source publication and dataset description did not supply a dataset-2a-specific committee, approval number or consent statement; their acquisition ethics details are not inferred from another dataset. The present analysis used coded records without participant recruitment, contact or re-identification.

Informed consent

The original Kumar study reported written informed consent before data collection. No dataset-2a-specific consent statement was identified in the reviewed BNCI source documents. The present study analyzed coded public records; no new participants were recruited or contacted and no additional consent was collected.

Research funding

This research received no external funding.

Conflict of interest

The authors declare no conflict of interest.

Author contributions

Ziyuan Zhu: Conceptualization; Methodology; Software; Formal analysis; Validation; Visualization; Writing – original draft; Writing – review & editing.

Sihao Liu assisted with code auditing and verification, supervised the correct execution of training, checked publication policies and organized project files.

Haoyang Yu assisted the first author with experimental design and code modification.

All authors critically reviewed the manuscript, approved the final version, consented to its submission to Journal of Neural Engineering, and agree to be accountable for all aspects of the work.

AI disclosure

OpenAI Codex assisted with code revision, data analysis and result checks, manuscript preparation, GitHub updates and file organization; OpenAI ChatGPT assisted with grammatical correction. The authors are responsible for verifying sources, scientific claims and the final manuscript. Validation included matching trial identities and labels, checking algebraic metric identities, replaying saved numerical outputs, verifying fitted-state invariance and reproducing saved tables; these checks are described in the Methods and supplementary materials. Tool-version identifiers were not recorded.

Data and software availability

Kumar2024 source EEG: NEMAR nm000177/v1.0.5, https://doi.org/10.82901/nemar.nm000177.v1.0.5, CC BY 4.0. BNCI2014-001 original recordings: https://bnci-horizon-2020.eu/database/data-sets/001-2014/, CC BY-ND 4.0; NEMAR nm000139/v1.0.3 (https://nemar.org/dataset/nm000139?v=v1.0.3) provides a versioned mirror with original-file hashes. The accompanying Supplementary Data and Code archive contains code, derived data, figure data, provenance and reproduction instructions. Its fixed scientific version is recorded at https://github.com/jackzhu119/few-shot-mi-eeg/tree/e5969e6ee6294206954b5a83c5258fdb2e143eda/manuscript/bmt_alignment_integrated_20261104. Current JNE manuscript materials are on branch `research/paper2-alignment-integrated-revision` of the same repository. Original project code is MIT; original derived data, figures and manuscript content are CC BY 4.0 under LICENSING.md. Original EEG and third-party materials retain their original terms.